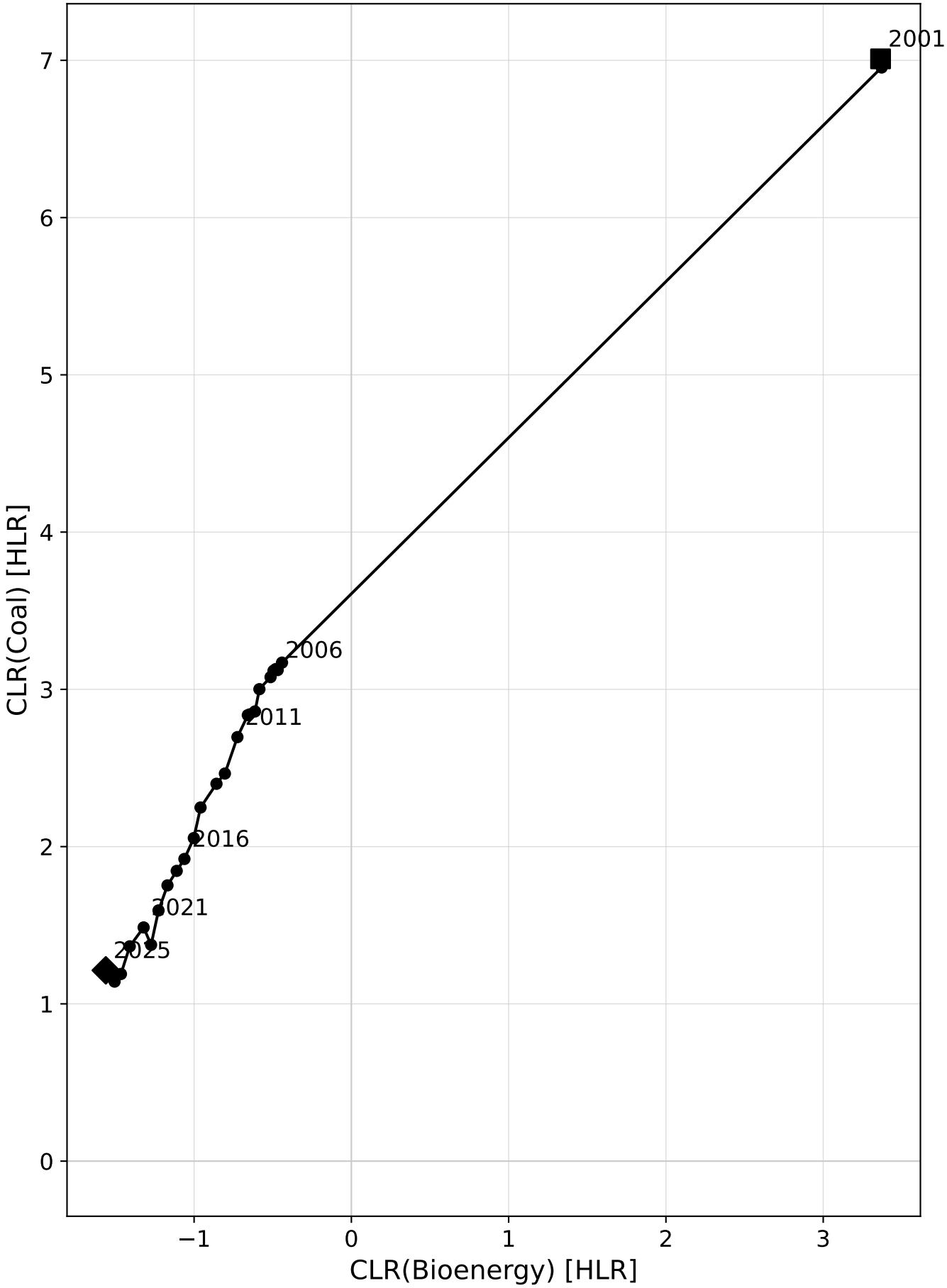
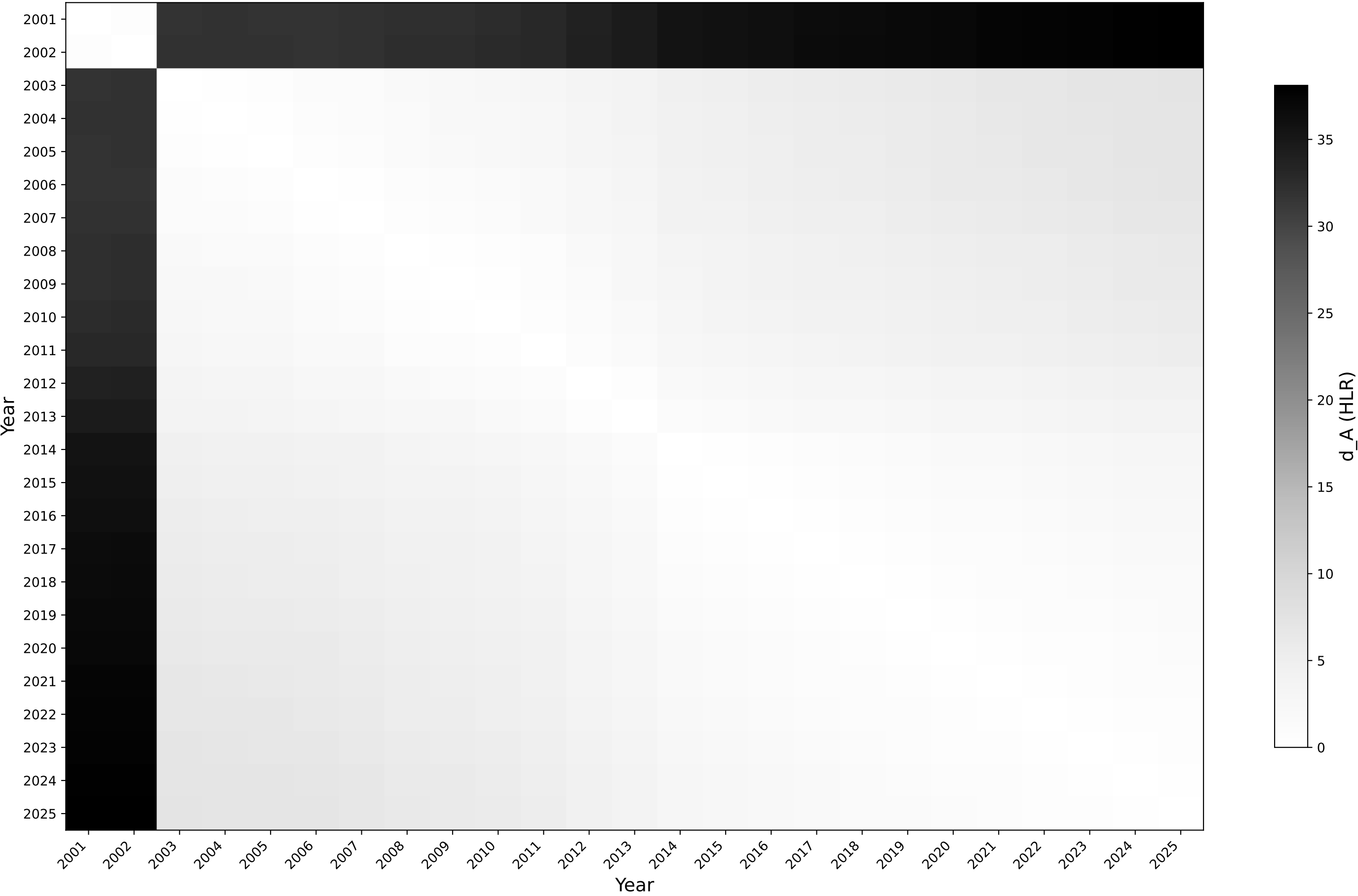


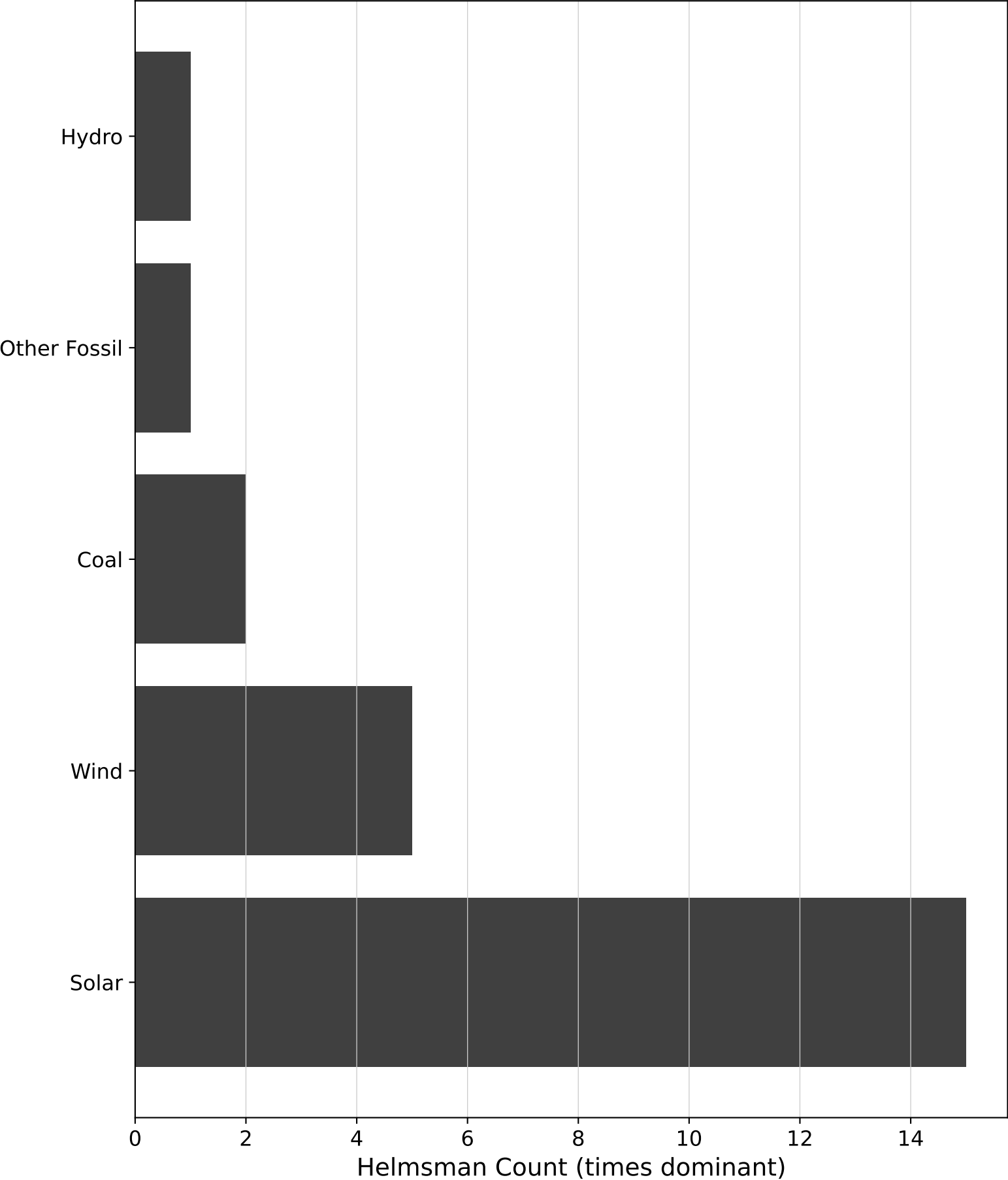
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



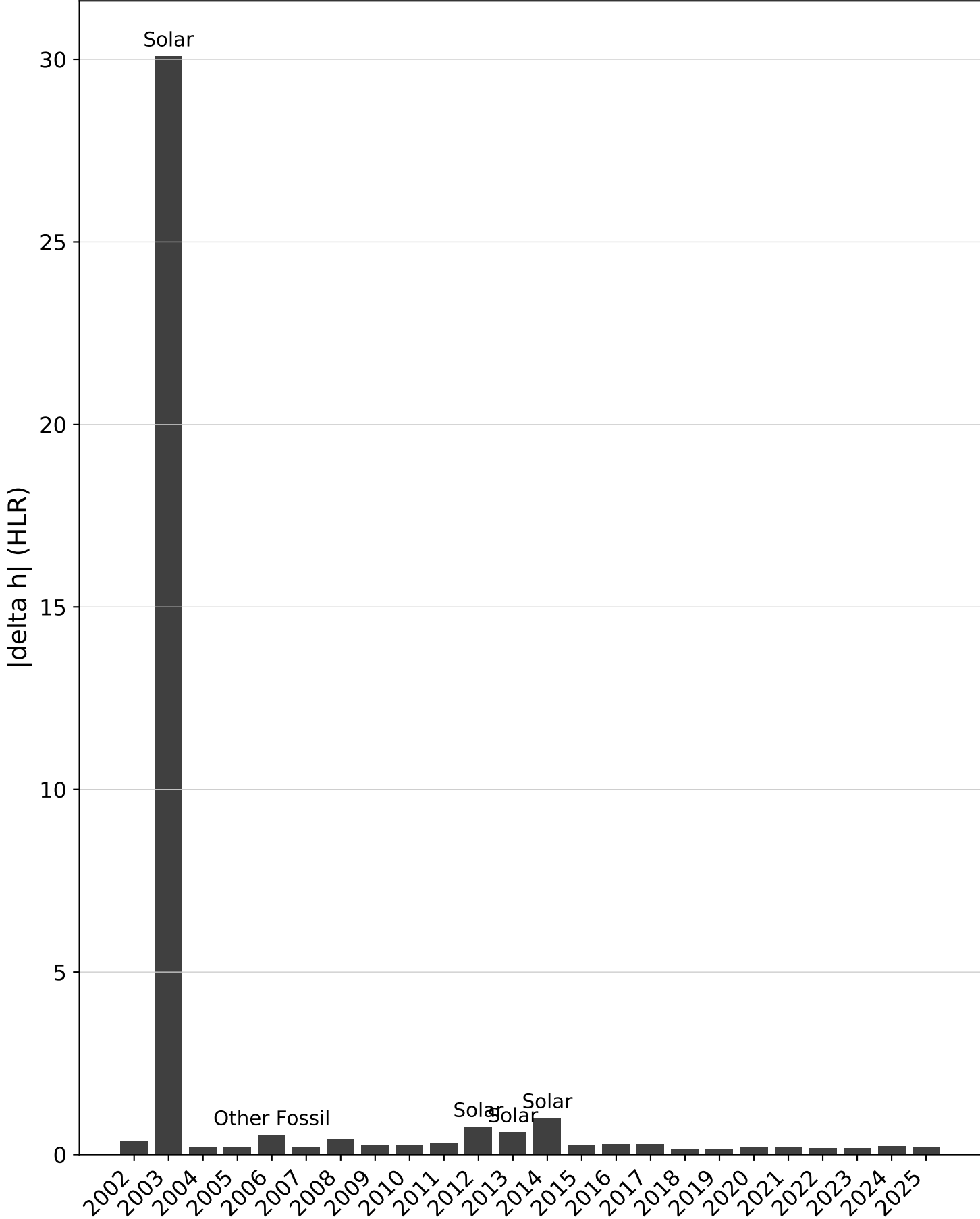
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



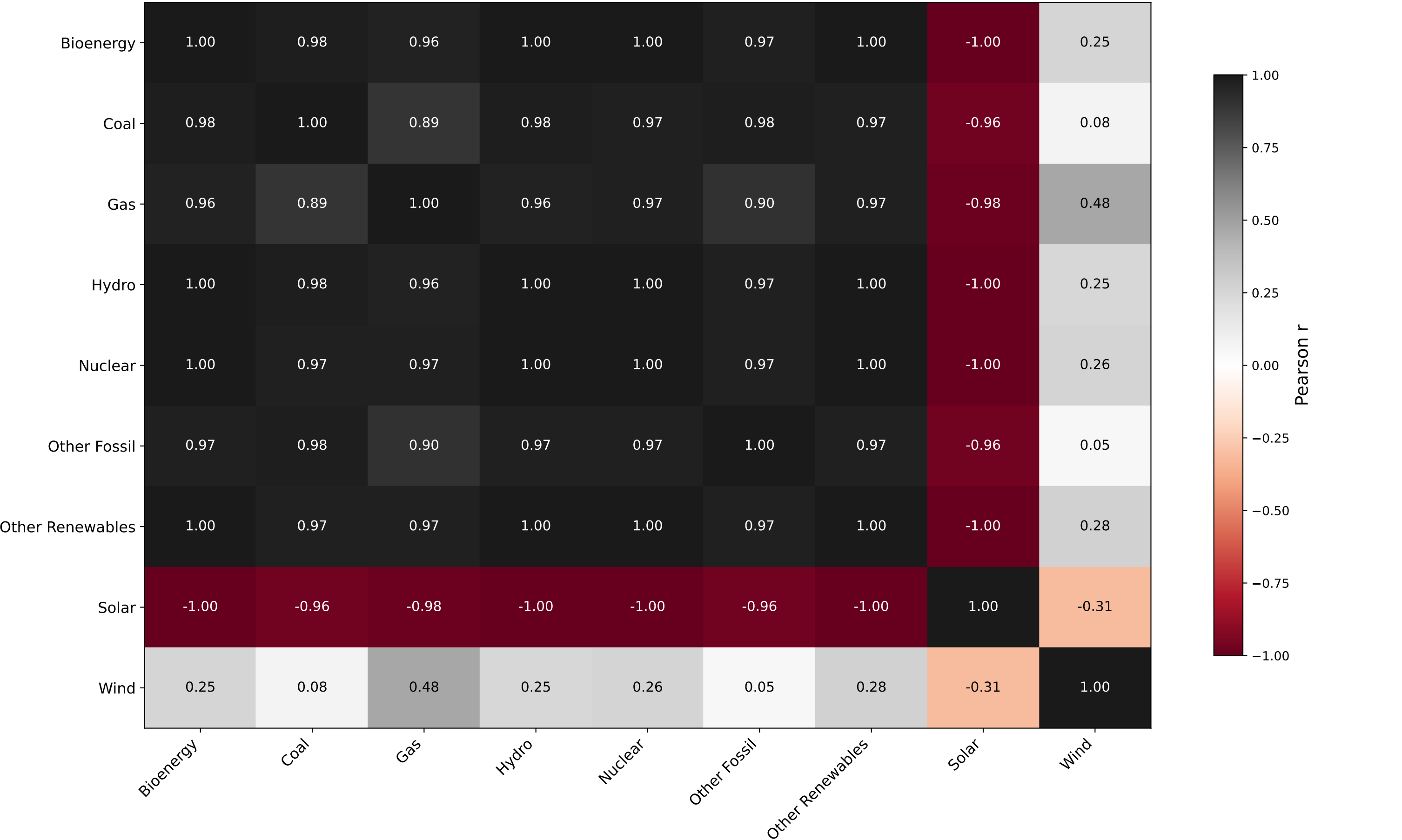
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2002	2025	38.1117	92.7	Solar
2	2001	2025	38.0885	93.0	Solar
3	2002	2024	37.9059	90.1	Solar
4	2001	2024	37.8834	90.4	Solar
5	2002	2023	37.6593	87.1	Solar
6	2001	2023	37.6357	87.4	Solar
7	2002	2022	37.4542	84.5	Solar
8	2001	2022	37.4295	84.7	Solar
9	2002	2021	37.2539	81.9	Solar
10	2001	2021	37.2288	82.1	Solar
11	2002	2020	37.0480	79.2	Solar
12	2001	2020	37.0230	79.4	Solar
13	2002	2019	36.8108	76.1	Solar
14	2001	2019	36.7835	76.3	Solar
15	2002	2018	36.6338	73.7	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2002	2003	2024	67.5061	5.31
2	2002	2003	2025	66.7205	5.24
3	2001	2003	2024	66.0803	5.31
4	2002	2003	2023	65.6965	5.48
5	2001	2003	2025	65.2449	5.23
6	2001	2003	2023	64.3308	5.47
7	2002	2003	2022	63.5557	5.64
8	2002	2004	2024	63.5441	5.43
9	2002	2004	2025	62.7167	5.35
10	2001	2003	2022	62.2329	5.64

STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2001	0.3884	Hs-3	895.1	0.00	0.0000	---	1000000000000000
2002	0.3837	Hs-3	895.5	0.79	0.5229	Wind	1000000000000000
2003	0.3791	Hs-3	53.1	34.99	31.9255	Solar	7300
2004	0.3761	Hs-3	51.9	1.69	0.2289	Wind	6933
2005	0.3756	Hs-3	51.9	1.86	0.2343	Wind	7336
2006	0.3799	Hs-3	52.3	5.62	0.7080	Other Fossil	8259
2007	0.3812	Hs-3	49.7	2.40	0.3509	Hydro	6791
2008	0.3765	Hs-3	45.5	4.71	0.6427	Wind	4721
2009	0.3535	Hs-3	44.8	2.94	0.3488	Wind	4427
2010	0.3552	Hs-3	42.2	2.55	0.3512	Solar	3372
2011	0.3350	Hs-3	38.9	3.64	0.4789	Solar	2247
2012	0.3260	Hs-3	32.3	6.57	0.8815	Solar	931
2013	0.3181	Hs-3	27.4	5.56	0.6909	Solar	449
2014	0.3061	Hs-3	21.8	10.67	1.0804	Solar	215
2015	0.2967	Hs-2	20.8	4.32	0.3662	Solar	214
2016	0.2824	Hs-2	20.1	4.91	0.3937	Solar	213
2017	0.2630	Hs-2	19.4	4.50	0.3596	Solar	212
2018	0.2641	Hs-2	19.2	3.08	0.2374	Solar	225
2019	0.2649	Hs-2	19.6	3.72	0.2893	Coal	233
2020	0.2591	Hs-2	19.2	4.45	0.3444	Coal	224
2021	0.2503	Hs-2	19.3	3.96	0.3033	Solar	227
2022	0.2466	Hs-2	19.3	3.51	0.2691	Solar	236
2023	0.2583	Hs-2	19.7	4.02	0.3132	Solar	228
2024	0.2562	Hs-2	20.8	3.13	0.2720	Solar	251
2025	0.2400	Hs-2	21.0	3.06	0.2448	Solar	257